1) a)

$\operatorname{cosech} x = \frac{2}{e^x - e^{-x}} = \frac{2e^x}{e^{2x} - 1}$

1) b)

$\operatorname{sech} x = \frac{2}{e^x + e^{-x}} = \frac{2e^x}{e^{2x} + 1}$

1) c)

$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{e^{2x} + 1}{e^{2x} - 1}$

2) a)

$\operatorname{cosech} 1 = \frac{2}{e^1 - e^{-1}} = 0.851$ (3 SF)

2) b)

$\operatorname{sech} 1 = \frac{2}{e^1 + e^{-1}} = 0.648$ (3 SF)

2) c)

$\operatorname{cosech}(-1) = \frac{2}{e^{-1} - e^1} = -0.851$ (3 SF)

2) d)

$\operatorname{sech}(-1) = \frac{2}{e^{-1} + e^1} = 0.648$ (3 SF)

3)

$$\operatorname{cosech}(-x)$$
$$\equiv \frac{2}{e^{-x} - e^{-(-x)}}$$
$$\equiv \frac{2}{e^{-x} - e^{x}}$$
$$\equiv \frac{2}{-(e^{x} - e^{-x})}$$
$$\equiv -\frac{2}{e^{x} - e^{-x}}$$
$$\equiv -\operatorname{cosech} x$$

4)

$\operatorname{sech}(-x)$

$\equiv \frac{2}{e^{-x}+e^{-(-x)}}$

$\equiv \frac{2}{e^{-x}+e^{x}}$

$\equiv \frac{2}{e^{x}+e^{-x}}$

$\equiv \operatorname{sech} x$

5) a)

$\operatorname{cosech}(\ln 3)$

$\equiv \frac{2}{e^{\ln 3}-e^{-\ln 3}}$

$\equiv \frac{2}{e^{\ln 3}-e^{\ln \frac{1}{3}}}$

$\equiv \frac{2}{3-\frac{1}{3}}$

$\equiv \frac{2}{\frac{8}{3}}$

$\equiv \frac{3}{4}$

5) b)

$\operatorname{sech}(\ln 6)$

$\equiv \frac{2}{e^{\ln 6}+e^{-\ln 6}}$

$\equiv \frac{2}{e^{\ln 6}+e^{\ln \frac{1}{6}}}$

$\equiv \frac{2}{6+\frac{1}{6}}$

$\equiv \frac{2}{\frac{37}{6}}$

$\equiv \frac{12}{37}$

5) c)

$\operatorname{cosech}\left(\ln\sqrt{5}\right)$

$\equiv \frac{2}{e^{\ln\sqrt{5}}-e^{-\ln\sqrt{5}}}$

$\equiv \frac{2}{e^{\ln\sqrt{5}}-e^{\ln\frac{1}{\sqrt{5}}}}$

$\equiv \frac{2}{\sqrt{5}-\frac{1}{\sqrt{5}}}$

$\equiv \frac{2\sqrt{5}}{5-1}$

$\equiv \frac{\sqrt{5}}{2}$

5) d)

$\operatorname{sech}(4\ln 3)$

$\equiv \frac{2}{e^{4\ln 3}+e^{-4\ln 3}}$

$\equiv \frac{2}{e^{\ln 81}+e^{\ln\frac{1}{81}}}$

$\equiv \frac{2}{81+\frac{1}{81}}$

$\equiv \frac{2}{\frac{6562}{81}}$

$\equiv \frac{81}{3281}$

6)

$\operatorname{cosech}(\ln k)$

$= \frac{2}{e^{\ln k}-e^{-\ln k}}$

$= \frac{2}{e^{\ln k}-e^{\ln\frac{1}{k}}}$

$= \frac{2}{k-\frac{1}{k}}$

$= \frac{2k}{k^2-1}$

7)

$\operatorname{sech}(3\ln k)$

$= \frac{2}{e^{3\ln k}+e^{-3\ln k}}$

$= \frac{2}{e^{\ln k^3}+e^{\ln\frac{1}{k^3}}}$

$= \frac{2}{k^3+\frac{1}{k^3}}$

$= \frac{2k^3}{k^6+1}$

8) a)

$f(1) = \operatorname{cosech}(2\ln 1) - \operatorname{sech}(2\ln 1)$

$= \operatorname{cosech} 0 - \operatorname{sech} 0$

$= \frac{1}{\sinh 0} - \frac{1}{\cosh 0}$

$= \frac{1}{0} - \frac{1}{1}$

So $f(x)$ is undefined when $x = 1$

8) b)

$f(x) = \operatorname{cosech}(2\ln x) - \operatorname{sech}(2\ln x)$

$= \frac{2}{e^{2\ln x}-e^{-2\ln x}} - \frac{2}{e^{2\ln x}+e^{-2\ln x}}$

$= \frac{2}{e^{\ln x^2}-e^{\ln x^{-2}}} - \frac{2}{e^{\ln x^2}+e^{\ln x^{-2}}}$

$= \frac{2}{x^2-x^{-2}} - \frac{2}{x^2+x^{-2}}$

$= \frac{2x^2}{x^4-1} - \frac{2x^2}{x^4+1}$

$= \frac{2x^2(x^4+1)-2x^2(x^4-1)}{(x^4-1)(x^4+1)}$

$= \frac{2x^6+2x^2-2x^6+2x^2}{(x^4-1)(x^4+1)}$

$= \frac{4x^2}{(x^4-1)(x^4+1)}$

9)

$$\coth x$$
$$\equiv \frac{\cosh x}{\sinh x}$$
$$\equiv \frac{\frac{e^x+e^{-x}}{2}}{\frac{e^x-e^{-x}}{2}}$$
$$\equiv \frac{e^x+e^{-x}}{e^x-e^{-x}}$$
$$\equiv \frac{e^{2x}+1}{e^{2x}-1}$$
$$\equiv \frac{e^{2x}+1}{(e^x+1)(e^x-1)}$$

10) a)

$$\coth(\ln 2)$$
$$\equiv \frac{e^{2\ln 2}+1}{e^{2\ln 2}-1}$$
$$\equiv \frac{e^{\ln 4}+1}{e^{\ln 4}-1}$$
$$\equiv \frac{4+1}{4-1}$$
$$\equiv \frac{5}{3}$$

10) b)

$$\coth\left(\ln\sqrt{2}\right)$$
$$\equiv \frac{e^{2\ln\sqrt{2}}+1}{e^{2\ln\sqrt{2}}-1}$$
$$\equiv \frac{e^{\ln 2}+1}{e^{\ln 2}-1}$$
$$\equiv \frac{2+1}{2-1}$$
$$\equiv 3$$

10) c)

$$\coth\left(\frac{1}{4}\ln 8\right)$$
$$\equiv \frac{e^{2\times\frac{1}{4}\ln 8}+1}{e^{2\times\frac{1}{4}\ln 8}-1}$$
$$\equiv \frac{e^{\frac{1}{2}\ln 8}+1}{e^{\frac{1}{2}\ln 8}-1}$$
$$\equiv \frac{e^{\ln\sqrt{8}}+1}{e^{\ln\sqrt{8}}-1}$$
$$\equiv \frac{\sqrt{8}+1}{\sqrt{8}-1}$$
$$\equiv \frac{2\sqrt{2}+1}{2\sqrt{2}-1}\times\frac{2\sqrt{2}+1}{2\sqrt{2}+1}$$
$$\equiv \frac{9+4\sqrt{2}}{7}$$

11)

$$\coth(-x)$$
$$\equiv \frac{e^{2(-x)}+1}{e^{2(-x)}-1}$$
$$\equiv \frac{e^{-2x}+1}{e^{-2x}-1}$$
$$\equiv \frac{1+e^{2x}}{1-e^{2x}}$$
$$\equiv \frac{e^{2x}+1}{-(e^{2x}-1)}$$
$$\equiv -\frac{e^{2x}+1}{e^{2x}-1}$$
$$\equiv -\coth x$$

12) a)

$$\coth(-\ln 3)$$
$$\equiv -\coth(\ln 3)$$
$$\equiv -\frac{e^{2\ln 3}+1}{e^{2\ln 3}-1}$$
$$\equiv -\frac{e^{\ln 9}+1}{e^{\ln 9}-1}$$
$$\equiv -\frac{9+1}{9-1}$$
$$\equiv -\frac{5}{4}$$

12) b)

$\coth(-5\ln 2)$

$\equiv -\coth(5\ln 2)$

$\equiv -\frac{e^{2\times 5\ln 2}+1}{e^{2\times 5\ln 2}-1}$

$\equiv -\frac{e^{10\ln 2}+1}{e^{10\ln 2}-1}$

$\equiv -\frac{e^{\ln 1024}+1}{e^{\ln 1024}-1}$

$\equiv -\frac{1024+1}{1024-1}$

$\equiv -\frac{1025}{1023}$

13) a)

$\operatorname{cosech} x = 1$

$\Rightarrow \sinh x = 1$

$\Rightarrow x = \ln\left(1+\sqrt{1^2+1}\right)$

$\Rightarrow x = \ln\left(1+\sqrt{2}\right)$

13) b)

$\operatorname{cosech} x = 2$

$\Rightarrow \sinh x = \frac{1}{2}$

$\Rightarrow x = \ln\left(\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2+1}\right)$

$\Rightarrow x = \ln\left(\frac{1}{2}+\sqrt{\frac{5}{4}}\right)$

$\Rightarrow x = \ln\left(\frac{1+\sqrt{5}}{2}\right)$

13) c)

$\operatorname{cosech} x = -\frac{1}{5}$

$\Rightarrow \sinh x = -5$

$\Rightarrow x = \ln\left(-5+\sqrt{(-5)^2+1}\right)$

$\Rightarrow x = \ln\left(-5+\sqrt{26}\right)$

14) a)

$\operatorname{sech} x = 1$

$\Rightarrow \cosh x = 1$

$\Rightarrow x = \ln\left(1 + \sqrt{1^2 - 1}\right)$

$\Rightarrow x = \ln 1$

$\Rightarrow x = 0$

14) b)

$\operatorname{sech} x = \frac{1}{2}$

$\Rightarrow \cosh x = 2$

$\Rightarrow x = \pm \ln\left(2 + \sqrt{2^2 - 1}\right)$

$\Rightarrow x = \pm \ln\left(2 + \sqrt{3}\right)$

$\Rightarrow x = \ln\left(2 \pm \sqrt{3}\right)$

14) c)

$\operatorname{sech} x = \frac{2}{3}$

$\Rightarrow \cosh x = \frac{3}{2}$

$\Rightarrow x = \pm \ln\left(\frac{3}{2} + \sqrt{\left(\frac{3}{2}\right)^2 - 1}\right)$

$\Rightarrow x = \pm \ln\left(\frac{3}{2} + \sqrt{\frac{5}{4}}\right)$

$\Rightarrow x = \pm \ln\left(\frac{3+\sqrt{5}}{2}\right)$

$\Rightarrow x = \ln\left(\frac{3\pm\sqrt{5}}{2}\right)$

15) a)

$\coth x = 2$

$\Rightarrow \tanh x = \frac{1}{2}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right)$

$\Rightarrow x = \frac{1}{2}\ln 3$

15) b)

$\coth x = \frac{3}{2}$

$\Rightarrow \tanh x = \frac{2}{3}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{2}{3}}{1-\frac{2}{3}}\right)$

$\Rightarrow x = \frac{1}{2}\ln 5$

15) c)

$\coth x = -\frac{5}{3}$

$\Rightarrow \tanh x = -\frac{3}{5}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{3}{5}\right)}{1-\left(-\frac{3}{5}\right)}\right)$

$\Rightarrow x = \frac{1}{2}\ln\frac{1}{4}$

$\Rightarrow x = -\frac{1}{2}\ln 4$

$\Rightarrow x = -\ln 2$

16) a)

$5\operatorname{sech} x - 2\tanh x = 5$

$\Rightarrow \frac{5}{\cosh x} - \frac{2\sinh x}{\cosh x} = 5$

$\Rightarrow 5 - 2\sinh x = 5\cosh x$

$\Rightarrow 2\sinh x + 5\cosh x = 5$

$\Rightarrow 2 \times \frac{e^x - e^{-x}}{2} + 5 \times \frac{e^x + e^{-x}}{2} = 5$

$\Rightarrow 2(e^x - e^{-x}) + 5(e^x + e^{-x}) = 10$

$\Rightarrow 2e^x - 2e^{-x} + 5e^x + 5e^{-x} = 10$

$\Rightarrow 7e^x + 3e^{-x} = 10$

$\Rightarrow 7e^{2x} + 3 = 10e^x$

$\Rightarrow 7e^{2x} - 10e^x + 3 = 0$

$\Rightarrow (7e^x - 3)(e^x - 1) = 0$

$\Rightarrow e^x = \frac{3}{7}$ or $e^x = 1$

$\Rightarrow x = \ln\frac{3}{7}$ or $x = 0$

16) b)

$8\operatorname{cosech} x - 3\coth x = 27$

$\Rightarrow \frac{8}{\sinh x} - \frac{3\cosh x}{\sinh x} = 27$

$\Rightarrow 8 - 3\cosh x = 27\sinh x$

$\Rightarrow 27\sinh x + 3\cosh x = 8$

$\Rightarrow 27 \times \frac{e^x - e^{-x}}{2} + 3 \times \frac{e^x + e^{-x}}{2} = 8$

$\Rightarrow 27(e^x - e^{-x}) + 3(e^x + e^{-x}) = 16$

$\Rightarrow 27e^x - 27e^{-x} + 3e^x + 3e^{-x} = 16$

$\Rightarrow 30e^x - 24e^{-x} = 16$

$\Rightarrow 15e^x - 12e^{-x} = 8$

$\Rightarrow 15e^{2x} - 12 = 8e^x$

$\Rightarrow 15e^{2x} - 8e^x - 12 = 0$

$\Rightarrow (3e^x + 2)(5e^x - 6) = 0$

$\Rightarrow e^x = -\frac{2}{3}$ or $e^x = \frac{6}{5}$

$e^x = -\frac{2}{3}$ has no real solutions

$e^x = \frac{6}{5} \Rightarrow x = \ln\frac{6}{5}$

16) c)

$2\tanh x - 5\coth x = 9$

$\Rightarrow 2\tanh x - \frac{5}{\tanh x} = 9$

$\Rightarrow 2\tanh^2 x - 5 = 9\tanh x$

$\Rightarrow 2\tanh^2 x - 9\tanh x - 5 = 0$

$\Rightarrow (2\tanh x + 1)(\tanh x - 5) = 0$

$\Rightarrow \tanh x = -\frac{1}{2}$ or $\tanh x = 5$

$\tanh x = 5$ has no real solutions

$\tanh x = -\frac{1}{2} \Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\frac{1}{3}$

17) a)

$\text{cosech}^2 x = 9$

$\Rightarrow \text{cosech}\, x = \pm 3$

If $\text{cosech}\, x = 3$

$\Rightarrow \sinh x = \frac{1}{3}$

$\Rightarrow x = \ln\left(\frac{1}{3} + \sqrt{\left(\frac{1}{3}\right)^2 + 1}\right) = \ln\left(\frac{1}{3} + \sqrt{\frac{10}{9}}\right) = \ln\left(\frac{1+\sqrt{10}}{3}\right)$

If $\text{cosech}\, x = -3$

$\Rightarrow \sinh x = -\frac{1}{3}$

$\Rightarrow x = \ln\left(-\frac{1}{3} + \sqrt{\left(-\frac{1}{3}\right)^2 + 1}\right) = \ln\left(-\frac{1}{3} + \sqrt{\frac{10}{9}}\right) = \ln\left(\frac{-1+\sqrt{10}}{3}\right)$

So $x = \ln\left(\frac{1+\sqrt{10}}{3}\right)$ or $x = \ln\left(\frac{-1+\sqrt{10}}{3}\right)$

17) b)

$\text{sech}^2 x = \frac{1}{16}$

$\Rightarrow \text{sech}\, x = \pm\frac{1}{4}$

If $\text{sech}\, x = \frac{1}{4}$

$\Rightarrow \cosh x = 4$

$\Rightarrow x = \ln\left(4 + \sqrt{4^2 - 1}\right) = \ln\left(4 + \sqrt{15}\right)$

If $\text{sech}\, x = -\frac{1}{4}$

$\Rightarrow \cosh x = -4$

$\Rightarrow x = \ln\left(-4 + \sqrt{(-4)^2 - 1}\right) = \ln\left(-4 + \sqrt{15}\right)$

So $x = \ln\left(4 + \sqrt{15}\right)$ or $x = \ln\left(-4 + \sqrt{15}\right)$

17) c)

$\coth^2 x = 3$

$\Rightarrow \coth x = \pm\sqrt{3}$

If $\coth x = \sqrt{3}$

$\Rightarrow \tanh x = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{\sqrt{3}}{3}}{1-\frac{\sqrt{3}}{3}}\right) = \frac{1}{2}\ln\left(\frac{3+\sqrt{3}}{3-\sqrt{3}}\right) = \frac{1}{2}\ln\left(2+\sqrt{3}\right)$

If $\coth x = -\sqrt{3}$

$\Rightarrow \tanh x = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{\sqrt{3}}{3}\right)}{1-\left(-\frac{\sqrt{3}}{3}\right)}\right) = \frac{1}{2}\ln\left(\frac{3-\sqrt{3}}{3+\sqrt{3}}\right) = \frac{1}{2}\ln\left(2-\sqrt{3}\right)$

So $x = \frac{1}{2}\ln\left(2+\sqrt{3}\right)$ or $x = \frac{1}{2}\ln\left(2-\sqrt{3}\right)$

18) a)

$\operatorname{cosech}^2 x + \operatorname{cosech} x - 20 = 0$

$\Rightarrow (\operatorname{cosech} x + 5)(\operatorname{cosech} x - 4) = 0$

$\Rightarrow \operatorname{cosech} x = -5$ or $\operatorname{cosech} x = 4$

If $\operatorname{cosech} x = -5$

$\Rightarrow \sinh x = -\frac{1}{5}$

$\Rightarrow x = \ln\left(-\frac{1}{5} + \sqrt{\left(-\frac{1}{5}\right)^2 + 1}\right) = \ln\left(-\frac{1}{5} + \sqrt{\frac{26}{25}}\right) = \ln\left(\frac{-1+\sqrt{26}}{5}\right)$

If $\operatorname{cosech} x = 4$

$\Rightarrow \sinh x = \frac{1}{4}$

$\Rightarrow x = \ln\left(\frac{1}{4} + \sqrt{\left(\frac{1}{4}\right)^2 + 1}\right) = \ln\left(\frac{1}{4} + \sqrt{\frac{17}{16}}\right) = \ln\left(\frac{1+\sqrt{17}}{4}\right)$

So $x = \ln\left(\frac{-1+\sqrt{26}}{5}\right)$ or $x = \ln\left(\frac{1+\sqrt{17}}{4}\right)$

18) b)

$10\operatorname{sech}^2 x - 9\operatorname{sech} x + 2 = 0$

$\Rightarrow (5\operatorname{sech} x - 2)(2\operatorname{sech} x - 1) = 0$

$\Rightarrow \operatorname{sech} x = \frac{2}{5}$ or $\operatorname{sech} x = \frac{1}{2}$

If $\operatorname{sech} x = \frac{2}{5}$

$\Rightarrow \cosh x = \frac{5}{2}$

$\Rightarrow x = \pm \ln\left(\frac{5}{2} + \sqrt{\left(\frac{5}{2}\right)^2 - 1}\right) = \pm \ln\left(\frac{5}{2} + \sqrt{\frac{21}{4}}\right) = \pm \ln\left(\frac{5+\sqrt{21}}{2}\right)$

$\Rightarrow x = \ln\left(\frac{5\pm\sqrt{21}}{2}\right)$

If $\operatorname{sech} x = \frac{1}{2}$

$\Rightarrow \cosh x = 2$

$\Rightarrow x = \pm \ln\left(2 + \sqrt{2^2 - 1}\right) = \pm \ln\left(2 + \sqrt{3}\right)$

$\Rightarrow x = \ln\left(2 \pm \sqrt{3}\right)$

So $x = \ln\left(\frac{5\pm\sqrt{21}}{2}\right)$ or $x = \ln\left(2 \pm \sqrt{3}\right)$

18) c)

$3\coth^2 x - 16\coth x - 64 = 0$

$\Rightarrow (3\coth x + 8)(\coth x - 8) = 0$

$\Rightarrow \coth x = -\frac{8}{3}$ or $\coth x = 8$

If $\coth x = -\frac{8}{3}$

$\Rightarrow \tanh x = -\frac{3}{8}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{3}{8}\right)}{1-\left(-\frac{3}{8}\right)}\right) = \frac{1}{2}\ln\frac{5}{11}$

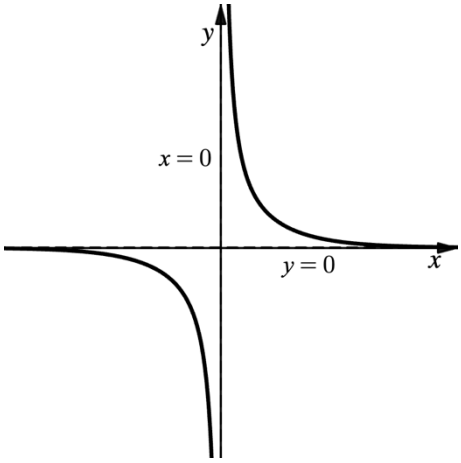
If $\coth x = 8$

$\Rightarrow \tanh x = \frac{1}{8}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{8}}{1-\frac{1}{8}}\right) = \frac{1}{2}\ln\frac{9}{7}$

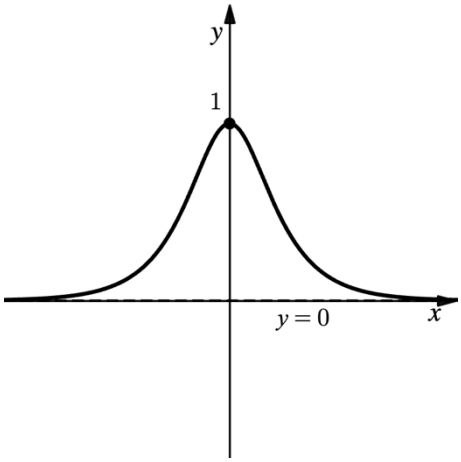
So $x = \frac{1}{2}\ln\frac{5}{11}$ or $x = \frac{1}{2}\ln\frac{9}{7}$

19) a)



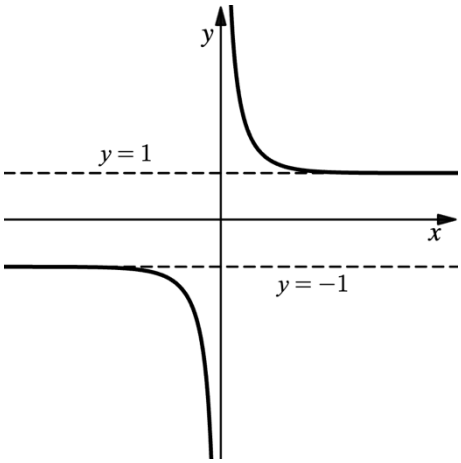
$y = \text{cosech}\, x$ is one-to-one

19) b)



$y = \text{sech}\, x$ is many-to-one

19) c)



$y = \coth x$ is one-to-one

20) a)

$y = \operatorname{cosech} x$

Domain: $\{x: x \in \mathbb{R}, x \neq 0\}$

Range: $\{y: y \in \mathbb{R}, y \neq 0\}$

20) b)

$y = \operatorname{sech} x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}, 0 < y \leq 1\}$

20) c)

$y = \coth x$

Domain: $\{x: x \in \mathbb{R}, x \neq 0\}$

Range: $\{y: y \in \mathbb{R}, y < -1\} \cup \{y: y \in \mathbb{R}, y > 1\}$

21) a)

$y = \operatorname{cosech} x$ where the domain is $\{x: x \in \mathbb{R}, x > 0\}$

Range: $\{y: y \in \mathbb{R}, y > 0\}$

21) b)

$y = \operatorname{sech} x$ where the domain is $\{x: x \in \mathbb{R}, x > 0\}$

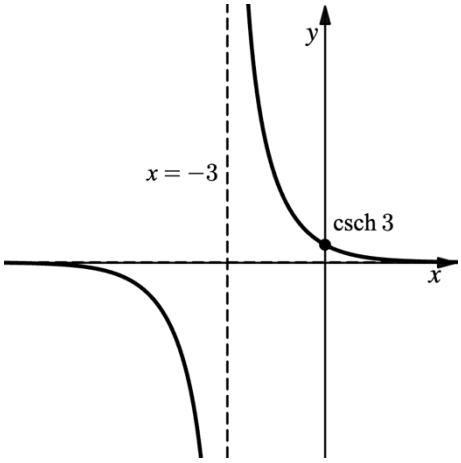
Range: $\{y: y \in \mathbb{R}, 0 < y < 1\}$

21) c)

$y = \coth x$ where the domain is $\{x: x \in \mathbb{R}, x > 0\}$

Range: $\{y: y \in \mathbb{R}, y > 1\}$

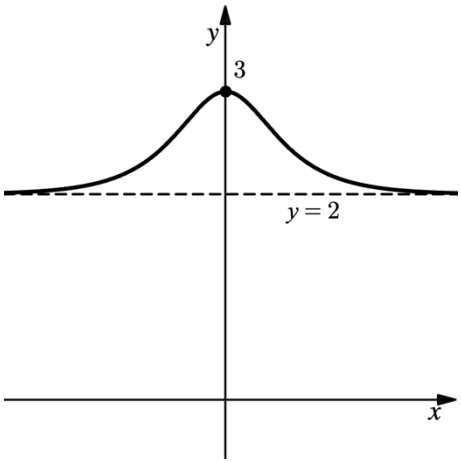
22) a)



Domain: $\{x: x \in \mathbb{R}, x \neq -3\}$

Range: $\{f(x): f(x) \in \mathbb{R}, f(x) \neq 0\}$

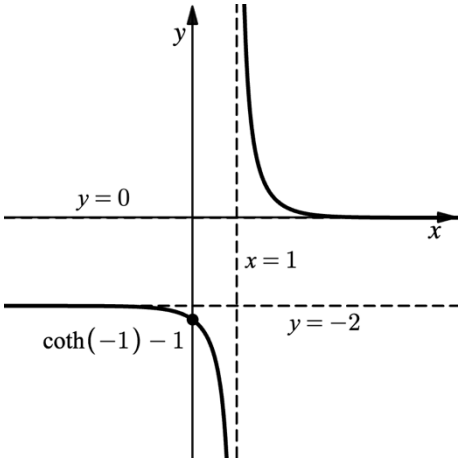
22) b)



Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{g(x): g(x) \in \mathbb{R}, 2 < g(x) \leq 3\}$

22) c)



Domain: $\{x: x \in \mathbb{R}, x \neq 1\}$

Range: $\{\mathrm{h}(x): \mathrm{h}(x) \in \mathbb{R}, \mathrm{h}(x) < -2\} \cup \{\mathrm{h}(x): \mathrm{h}(x) \in \mathbb{R}, \mathrm{h}(x) > 0\}$

23) a)

$\mathrm{f}(x) = \operatorname{cosech}(x - 2) + 2$

Domain: $\{x: x \in \mathbb{R}, x \neq 2\}$

Range: $\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \neq 2\}$

23) b)

$y = 2 - \operatorname{cosech} x$

Domain: $\{x: x \in \mathbb{R}, x \neq 0\}$

Range: $\{y: y \in \mathbb{R}, y \neq 2\}$

23) c)

$\mathrm{g}(x) = 3\operatorname{sech}(x + 3)$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{\mathrm{g}(x): \mathrm{g}(x) \in \mathbb{R}, 0 < \mathrm{g}(x) \leq 3\}$

23) d)

$y = 4 + 3\operatorname{sech} x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}, 4 < y \leq 7\}$

23) e)

$\mathrm{h}(x) = \coth(x+4) - 3$

Domain: $\{x: x \in \mathbb{R}, x \neq -4\}$

Range: $\{\mathrm{h}(x): \mathrm{h}(x) \in \mathbb{R}, \mathrm{h}(x) < -4\} \cup \{\mathrm{h}(x): \mathrm{h}(x) \in \mathbb{R}, \mathrm{h}(x) > -2\}$

23) f)

$y = 2 - \frac{1}{3}\coth x$

Domain: $\{x: x \in \mathbb{R}, x \neq 0\}$

Range: $\left\{y: y \in \mathbb{R}, y < \frac{5}{3}\right\} \cup \left\{y: y \in \mathbb{R}, y > \frac{7}{3}\right\}$

24) a)

$y = \mathrm{cosech}^2 x$

Domain: $\{x: x \in \mathbb{R}, x \neq 0\}$

Range: $\{y: y \in \mathbb{R}, y > 0\}$

24) b)

$y = \mathrm{sech}^2 x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}, 0 < y \leq 1\}$

24) c)

$y = \coth^2 x$

Domain: $\{x: x \in \mathbb{R}, x \neq 0\}$

Range: $\{y: y \in \mathbb{R}, y > 1\}$

25) a)

$\mathrm{cosech}\, x + \mathrm{sech}\, x$

$\equiv \frac{2}{\mathrm{e}^x - \mathrm{e}^{-x}} + \frac{2}{\mathrm{e}^x + \mathrm{e}^{-x}}$

$\equiv \frac{2\mathrm{e}^x + 2\mathrm{e}^{-x} + 2\mathrm{e}^x - 2\mathrm{e}^{-x}}{\mathrm{e}^{2x} - \mathrm{e}^{-2x}}$

$\equiv \frac{4\mathrm{e}^x}{\mathrm{e}^{2x} - \mathrm{e}^{-2x}}$

$\equiv \frac{4\mathrm{e}^{3x}}{\mathrm{e}^{4x} - 1}$

25) b)

$e^{3x} > 1$ for all $x > 0$

So $\frac{4e^{3x}}{e^{4x}-1} > \frac{4}{e^{4x}-1}$

$\therefore \operatorname{cosech} x + \operatorname{sech} x > \frac{4}{e^{4x}-1}$

26)

$\coth x - \tanh x$

$\equiv \frac{\cosh x}{\sinh x} - \frac{\sinh x}{\cosh x}$

$\equiv \frac{\frac{e^x+e^{-x}}{2}}{\frac{e^x-e^{-x}}{2}} - \frac{\frac{e^x-e^{-x}}{2}}{\frac{e^x+e^{-x}}{2}}$

$\equiv \frac{e^x+e^{-x}}{e^x-e^{-x}} - \frac{e^x-e^{-x}}{e^x+e^{-x}}$

$\equiv \frac{(e^x+e^{-x})^2-(e^x-e^{-x})^2}{(e^x-e^{-x})(e^x+e^{-x})}$

$\equiv \frac{(e^{2x}+2+e^{-2x})-(e^{2x}-2+e^{-2x})}{e^{2x}-e^{-2x}}$

$\equiv \frac{4}{e^{2x}-e^{-2x}}$

$\equiv \frac{4e^{2x}}{e^{4x}-1}$

Now $\frac{4e^{2x}}{e^{4x}-1} > \frac{4e^{2x}}{e^{4x}} \equiv 4e^{-2x}$

$\therefore \coth x - \tanh x > 4e^{-2x}$ for $x > 0$

27)

$\mathrm{f}(x) = \operatorname{cosech} x + 8\operatorname{sech} x$

$\mathrm{f}'(x) = -\operatorname{cosech} x \coth x - 8\operatorname{sech} x \tanh x$

$= -\frac{1}{\sinh x} \times \frac{\cosh x}{\sinh x} - 8 \times \frac{1}{\cosh x} \times \frac{\sinh x}{\cosh x}$

$= -\frac{\cosh x}{\sinh^2 x} - \frac{8\sinh x}{\cosh^2 x}$

$= \frac{-\cosh^3 x - 8\sinh^3 x}{\sinh^2 x \cosh^2 x}$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\Rightarrow \frac{-\cosh^3 x - 8\sinh^3 x}{\sinh^2 x \cosh^2 x} = 0$

$\Rightarrow -\cosh^3 x - 8\sinh^3 x = 0$

$\Rightarrow -1 - 8\tanh^3 x = 0$

$\Rightarrow \tanh^3 x = -\frac{1}{8}$

$\Rightarrow \tanh x = -\frac{1}{2}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\left(\frac{\frac{1}{2}}{\frac{3}{2}}\right) = \frac{1}{2}\ln\frac{1}{3}$

$\mathrm{f}\left(\frac{1}{2}\ln\frac{1}{3}\right) = \operatorname{cosech}\left(\frac{1}{2}\ln\frac{1}{3}\right) + 8\operatorname{sech}\left(\frac{1}{2}\ln\frac{1}{3}\right)$

$= \frac{2}{e^{\frac{1}{2}\ln\frac{1}{3}} - e^{-\frac{1}{2}\ln\frac{1}{3}}} + 8 \times \frac{2}{e^{\frac{1}{2}\ln\frac{1}{3}} + e^{-\frac{1}{2}\ln\frac{1}{3}}}$

$= \frac{2}{e^{\ln\left(\frac{1}{\sqrt{3}}\right)} - e^{\ln(\sqrt{3})}} + \frac{16}{e^{\ln\left(\frac{1}{\sqrt{3}}\right)} + e^{\ln(\sqrt{3})}}$

$= \frac{2}{\frac{1}{\sqrt{3}} - \sqrt{3}} + \frac{16}{\frac{1}{\sqrt{3}} + \sqrt{3}}$

$= \frac{2\sqrt{3}}{1-3} + \frac{16\sqrt{3}}{1+3} = -\sqrt{3} + 4\sqrt{3} = 3\sqrt{3}$

Range is $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 3\sqrt{3}\right\}$

28)

$f(x) = \cosh x + \coth x$

$f'(x) = \sinh x - \operatorname{cosech}^2 x$

$= \sinh x - \frac{1}{\sinh^2 x}$

$= \frac{\sinh^3 x - 1}{\sinh^2 x}$

Stationary points exist when $f'(x) = 0$

$\Rightarrow \frac{\sinh^3 x - 1}{\sinh^2 x} = 0$

$\Rightarrow \sinh^3 x - 1 = 0$

$\Rightarrow \sinh^3 x = 1$

$\Rightarrow \sinh x = 1$

$\Rightarrow x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

$f\left(\ln\left(1 + \sqrt{2}\right)\right) = \cosh\left(\ln\left(1 + \sqrt{2}\right)\right) + \coth\left(\ln\left(1 + \sqrt{2}\right)\right)$

$= \frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{2} + \frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}$

$= \frac{e^{\ln(1+\sqrt{2})} + e^{\ln(\sqrt{2}-1)}}{2} + \frac{e^{\ln(1+\sqrt{2})} + e^{\ln(\sqrt{2}-1)}}{e^{\ln(1+\sqrt{2})} - e^{\ln(\sqrt{2}-1)}}$

$= \frac{1+\sqrt{2}+\sqrt{2}-1}{2} + \frac{1+\sqrt{2}+\sqrt{2}-1}{1+\sqrt{2}-(\sqrt{2}-1)}$

$= \frac{2\sqrt{2}}{2} + \frac{2\sqrt{2}}{2}$

$= \sqrt{2} + \sqrt{2} = 2\sqrt{2}$

Range is $f(x) \in \left[2\sqrt{2}, \infty\right)$